

Prob 1 - Polynomial interpolation

$$\begin{array}{ccccc} x_i & -2 & -1 & 0 & 1 & 2 \\ y_i & \frac{1}{2} & 5 & \frac{5}{2} & -1 & \frac{1}{2} \end{array}$$

Using Lagrange interpolation method

$$l_i = \prod_{\substack{n=0 \\ n \neq i}}^n \frac{x - x_n}{x_i - x_n}$$

$$l_0 = \left(\frac{x - (-1)}{-2 - (-1)} \right) \left(\frac{x - 0}{-2 - 0} \right) \left(\frac{x - 1}{-2 - 1} \right) \left(\frac{x - 2}{-2 - 2} \right) = \frac{(x+1)x(x-1)(x-2)}{-1 \cdot (-2) \cdot (-3) \cdot (-4)} = \frac{x^4}{24} - \frac{x^3}{12} - \frac{x^2}{24} + \frac{x}{12}$$

$$l_1 = \left(\frac{x - (-2)}{-1 - (-2)} \right) \left(\frac{x - 0}{-1 - 0} \right) \left(\frac{x - 1}{-1 - 1} \right) \left(\frac{x - 2}{-1 - 2} \right) = \frac{(x+2)x(x-1)(x-2)}{1 \cdot (-1) \cdot (-2) \cdot (-3)} = -\frac{x^4}{6} + \frac{x^3}{6} + \frac{2x^2}{3} - \frac{2x}{3}$$

$$l_2 = \left(\frac{x - (-2)}{0 - (-2)} \right) \left(\frac{x - (-1)}{0 - (-1)} \right) \left(\frac{x - 1}{0 - 1} \right) \left(\frac{x - 2}{0 - 2} \right) = \frac{(x+2)(x+1)(x-1)(x-2)}{2 \cdot 1 \cdot (-1) \cdot (-2)} = \frac{x^4}{4} - \frac{5x^2}{4} + 1$$

$$l_3 = \left(\frac{x - (-2)}{1 - (-2)} \right) \left(\frac{x - (-1)}{1 - (-1)} \right) \left(\frac{x - 0}{1 - 0} \right) \left(\frac{x - 2}{1 - 2} \right) = \frac{(x+2)(x+1)x(x-2)}{3 \cdot 2 \cdot 1 \cdot (-1)} = -\frac{x^4}{6} - \frac{x^3}{6} + \frac{2x^2}{3} + \frac{2x}{3}$$

$$l_4 = \left(\frac{x - (-2)}{2 - (-2)} \right) \left(\frac{x - (-1)}{2 - (-1)} \right) \left(\frac{x - 0}{2 - 0} \right) \left(\frac{x - 1}{2 - 1} \right) = \frac{(x+2)(x+1)x(x-1)}{4 \cdot 3 \cdot 2 \cdot 1} = \frac{x^4}{24} + \frac{x^3}{12} - \frac{x^2}{24} - \frac{x}{12}$$

$$h(x) = \sum l_i y_i = \frac{1}{2} l_0 + 5 l_1 + \frac{5}{2} l_2 - l_3 + \frac{1}{2} l_4$$

Prob 2 - Fixed-point and Newton iterations

$$x = g(x) \quad , \quad g(x) = -1.93 \ln(x) + 15.9$$

a) $g'(x) = -\frac{1.93}{x}$

We can see that $g'(x) < 0 \quad \forall x > 0$, hence $g(x)$ is a decreasing function.

b) Since $g(x)$ is a decreasing function, we know that the maximum and minimum values of $g(x)$, respectively lies in the start and the end of the interval.

max: $g(e) = \underline{13.97}$

min: $g(e^3) = \underline{10.11}$

c) $|g'(x)| < 1 \Rightarrow -1 < g'(x) < 1 \Rightarrow g'(e) = 0.71 \wedge g'(e^3) = 0.096$

d) Same argument as b), "since $g(x)$ is a decreasing function" we know that the values at the start- and endpoint of the interval are the max and min values of the function. We calculated these values in b). Assumption 1) $g(x) \in [a, b] \quad \forall x \in [a, b]$ of fixed point OK.

we showed Assumption a) $g'(x) \leq L < 1 \quad \forall x \in [a, b]$ in c). Since both assumptions are OK,

we know that there exists a unique fixed point whom all x_0 will converge to.

e) $x_{k+1} = g(x_k)$

$$x_0 = e^2$$

$$x_1 = g(e^2) = 12,04$$

f) From the def of fixed point iteration, we know that $F(x)$ can be written as

$$F(x) = g(x) - x$$

$$F(x) = -1.93 \ln(x) + 15.9 - x$$

$$F(x) = -\frac{1.93}{x} - 1$$

$$x_{k+1} = x_k - \frac{F(x_k)}{F'(x_k)}$$

$$x_0 = e^2$$

$$x_1 = e^2 - \frac{F(e^2)}{F'(e^2)} = 11,08$$

Prob 3 - Runge's method

$$\begin{array}{c|cc} 0 & 0 & 0 \\ \frac{2}{3} & \frac{2}{3} & 0 \\ \hline & \frac{1}{4} & \frac{3}{4} \end{array} \quad y'(t) = -by(t) = F(t, y), \quad y(0) = 1$$

a) This RK-method has 2 stages

b) $k_1 = F(x_n, y_n) = -by_n$

$$k_2 = F(x_n + \frac{2}{3}h, y_n + h \sum_{i,j} a_{ij} k_j) = k_1$$

$$k_2 = F(x_n + \frac{2}{3}h, y_n + \frac{2}{3}h F(x_n, y_n)) = F(x_n + \frac{2}{3}h, y_n + \frac{2}{3}h(-by_n)) = -b(y_n - 4hy_n)$$

$$y_{n+1} = y_n + h \sum b_i k_i$$

$$= y_n + h(\frac{1}{4} k_1 + \frac{3}{4} k_2)$$

$$= y_n + \frac{h}{4}(-by_n) + 3(-b(y_n - 4hy_n))$$

$$= y_n - \frac{b}{4} h y_n - \frac{18}{4} h y_n + \frac{72}{4} h^2 y_n$$

$$= y_n - 6y_n h + 18h^2 y_n$$

$$= y_n (1 - 6h + 18h^2) \Rightarrow R(h) = 1 - 6h + 18h^2$$

c) Stability means that y_n does not tend to infinity for $n \rightarrow \infty$

since $y_n = [R(h)]^n y(0)$ and

$$R(h) = 1 - 6h + 18h^2 > 0 \quad \forall h \in \mathbb{R}$$

all we need to guarantee is $R(h) \leq 1$

$$1 - 6h + 18h^2 \leq 1 \Leftrightarrow 18h^2 - 6h \leq 0 \Leftrightarrow 6h(3h - 1) \leq 0 \Leftrightarrow 3h - 1 \leq 0 \Leftrightarrow \underline{h \leq \frac{1}{3}}$$

Prob 4 - Laplace transform

a) $f(t) = t^2 e^{-4t}$

We have that $\mathcal{L}(e^{at} f(t)) = F(s-a)$, which gives

$$F(s) = \frac{2}{(s+4)^3}$$

$$e^{at} \quad at e^{at} \quad at^2 e^{at}$$

b) $F(s) = \frac{s+a}{(s-a)^3} \Rightarrow f(t) = e^{at} (1 + (a+a)t)$

$$F(s) = \frac{s-a}{(s-a)^3} + \frac{a+a}{(s-a)^2} = \frac{1}{(s-a)} + \frac{2}{(s-a)^2} + \frac{a}{(s-a)^3}$$

$$F(s-a) = \mathcal{L}(e^{at} f(t)) \Rightarrow \underline{e^{2t} (1 + 2t + at)}$$

c) $y'' - 4y' + 4y = 0, \quad y'(0) = y(0) = 1$

$$(s^2 F(s) - s f(0) - f'(0)) - 4(s F(s) - f(0)) + 4F(s) = 0$$

$$s^2 F(s) - s - 1 - 4s F(s) + 4 + 4F(s) = 0$$

$$F(s)(s^2 - 4s + 4) = s - 3$$

$$F(s) = \frac{s-3}{(s-2)(s-2)} = \frac{s-3}{(s-2)^2}$$

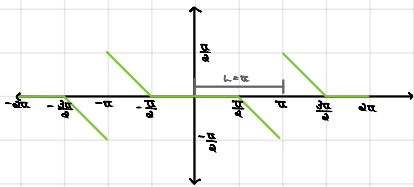
Using the result from b) with $a = -2$

$$f(t) = \underline{e^{-2t} (1-t)}$$

Prob 5 - Fourier Series

$$f(x) = \begin{cases} 0, & 0 \leq x < \frac{\pi}{2} \\ \frac{\pi}{2} - x, & \frac{\pi}{2} \leq x \leq \pi \end{cases}$$

a)



$$x_0 = \pi, x_1 = \frac{\pi}{2}$$

$$c) \lim_{x \rightarrow x_0^-} g(x) = -\frac{1}{2}, \lim_{x \rightarrow x_1^-} g(x) = 0$$

$$\lim_{x \rightarrow x_0^+} g(x) = \frac{1}{2}, \lim_{x \rightarrow x_1^+} g(x) = 0$$

b) We know that $f(x)$ is odd $\Rightarrow a_0 = a_n = 0$. Still need to compute b_n .

$$b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin(nx) dx, L = \pi$$

$$= \frac{2}{\pi} \int_{\frac{\pi}{2}}^{\pi} (\frac{\pi}{2} - x) \sin(nx) dx$$

$$= \frac{2}{\pi} \int_{\frac{\pi}{2}}^{\pi} \sin(nx) dx - \frac{2}{\pi} \int_{\frac{\pi}{2}}^{\pi} x \sin(nx) dx$$

$$= -\frac{1}{n} [\cos(nx)]_{\frac{\pi}{2}}^{\pi} + \frac{2}{n\pi} [x \cos(nx) + \int \cos(nx) dx]_{\frac{\pi}{2}}^{\pi}$$

$$= -\frac{1}{n} [\cos(nx)]_{\frac{\pi}{2}}^{\pi} + \frac{2}{n\pi} [x \cos(nx) + \frac{\sin(nx)}{n}]_{\frac{\pi}{2}}^{\pi}$$

$$= -\frac{1}{n} (\cos(n\pi) - \cos(\frac{n\pi}{2})) + \frac{2}{n\pi} (\pi \cos(n\pi) - 0 - (\frac{\pi}{2} \cos(\frac{n\pi}{2}) + \frac{\sin(\frac{n\pi}{2})}{n}))$$

$$= -\frac{(-1)^n}{n} + \frac{2(-1)^n}{n} - \frac{2\sin(\frac{n\pi}{2})}{\pi n^2}$$

$$= \frac{(-1)^n}{n} - \frac{2\sin(\frac{n\pi}{2})}{\pi n^2}$$

$$b_n = \begin{cases} \frac{(-1)^n}{n} - \frac{2(-1)^n}{\pi n^2}, & n \text{ is odd} \\ \frac{(-1)^n}{n}, & n \text{ is even} \end{cases}$$

$$f(x) \sim \sum b_n \sin(nx)$$

$$= \sum_{n=1}^{\infty} \frac{(-1)^n}{n} \sin(nx) + \frac{2}{\pi} \sum_{k=0}^{\infty} \frac{(-1)^{2k+1}}{(2k+1)^2} \sin((2k+1)x)$$

Prob 6 - Fourier Transform

The Fourier transform is given by

$$\hat{f}(\omega) = \mathcal{F}(f(x)) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-i\omega x} dx$$

With $f(x) = \cos(\lambda x)$, $x \in [-L, L]$, we get

$$\hat{f}(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-L}^L \cos(\lambda x) e^{-i\omega x} dx, \quad \cos(\lambda x) = \frac{e^{i\lambda x} + e^{-i\lambda x}}{2}$$

$$= \frac{1}{2\sqrt{2\pi}} \int_{-L}^L (e^{i\lambda x} + e^{-i\lambda x}) e^{-i\omega x} dx$$

$$= \frac{1}{2\sqrt{2\pi}} \int_{-L}^L e^{i(x(\lambda-\omega))} + e^{-i(x(\lambda+\omega))} dx$$

$$\begin{aligned}
&= \frac{1}{2\sqrt{2\pi}} \left[\frac{e^{\frac{i(L(\lambda-\omega))}{2}}}{i(\lambda-\omega)} - \frac{e^{\frac{-i(L(\lambda-\omega))}{2}}}{i(\lambda+\omega)} \right] \\
&= \frac{1}{2\sqrt{2\pi}} \left(\frac{e^{\frac{i(L(\lambda-\omega))}{2}}}{i(\lambda-\omega)} - \frac{e^{\frac{-i(L(\lambda-\omega))}{2}}}{i(\lambda+\omega)} - \left(\frac{e^{\frac{-i(L(\lambda-\omega))}{2}}}{i(\lambda-\omega)} - \frac{e^{\frac{i(L(\lambda-\omega))}{2}}}{i(\lambda+\omega)} \right) \right) \\
&= \frac{1}{2\sqrt{2\pi}} \left(\frac{e^{\frac{i(L(\lambda-\omega))}{2}} - e^{\frac{-i(L(\lambda-\omega))}{2}}}{i(\lambda-\omega)} + \frac{e^{\frac{-i(L(\lambda-\omega))}{2}} - e^{\frac{i(L(\lambda-\omega))}{2}}}{i(\lambda+\omega)} \right) \\
&= \frac{1}{2\sqrt{2\pi}} \left(\frac{\sin(L(\lambda-\omega))}{i(\lambda-\omega)} + \frac{\sin(L(\lambda+\omega))}{i(\lambda+\omega)} \right)
\end{aligned}$$

Prob 7 - DFT

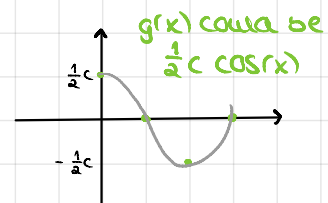
$$\vec{f} = (1/2, 1, 1/2, 0) \in \mathbb{R}^4, \quad \hat{f} = \mathcal{F}_4 \vec{f}$$

$$a) \quad \mathcal{F}_4 = \begin{bmatrix} e^{\frac{0 \cdot 0 \cdot 2\pi i}{4}} & e^{\frac{0 \cdot 1 \cdot 2\pi i}{4}} & e^{\frac{0 \cdot 2 \cdot 2\pi i}{4}} & e^{\frac{0 \cdot 3 \cdot 2\pi i}{4}} \\ e^{\frac{1 \cdot 0 \cdot 2\pi i}{4}} & e^{\frac{1 \cdot 1 \cdot 2\pi i}{4}} & e^{\frac{1 \cdot 2 \cdot 2\pi i}{4}} & e^{\frac{1 \cdot 3 \cdot 2\pi i}{4}} \\ e^{\frac{2 \cdot 0 \cdot 2\pi i}{4}} & e^{\frac{2 \cdot 1 \cdot 2\pi i}{4}} & e^{\frac{2 \cdot 2 \cdot 2\pi i}{4}} & e^{\frac{2 \cdot 3 \cdot 2\pi i}{4}} \\ e^{\frac{3 \cdot 0 \cdot 2\pi i}{4}} & e^{\frac{3 \cdot 1 \cdot 2\pi i}{4}} & e^{\frac{3 \cdot 2 \cdot 2\pi i}{4}} & e^{\frac{3 \cdot 3 \cdot 2\pi i}{4}} \end{bmatrix} = \begin{bmatrix} e^0 & e^0 & e^0 & e^0 \\ e^0 & e^i & e^{2i} & e^{3i} \\ e^0 & e^{2i} & e^{4i} & e^{6i} \\ e^0 & e^{3i} & e^{6i} & e^{9i} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -i & -1 & i \\ 1 & -1 & 1 & -1 \\ 1 & i & -1 & -i \end{bmatrix}$$

$$b) \quad \hat{f} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -i & -1 & i \\ 1 & -1 & 1 & -1 \\ 1 & i & -1 & -i \end{bmatrix} \begin{bmatrix} 1/2 \\ 1 \\ 1/2 \\ 0 \end{bmatrix} = \begin{bmatrix} 1/2 + 1 + 1/2 + 0 \\ 1/2 - i - 1/2 + 0 \\ 1/2 - 1 + 1/2 + 0 \\ 1/2 + i - 1/2 + 0 \end{bmatrix} = \begin{bmatrix} 2 \\ -i \\ 0 \\ i \end{bmatrix}$$

$$c) \quad \vec{g} \Rightarrow \hat{g} = (\hat{g}_0, \hat{g}_1, \hat{g}_2, \hat{g}_3) = (0, c, 0, c)$$

$$g = \mathcal{F}_4^{-1} \hat{g} = \frac{1}{4} \overline{\mathcal{F}_4} \hat{g} = \frac{1}{4} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & i \end{bmatrix} \begin{bmatrix} 0 \\ c \\ 0 \\ c \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 2c \\ 0 \\ -2c \\ 0 \end{bmatrix}$$



$$d) \quad n = \mathcal{F}_8^{-1} \hat{n}, \quad \hat{n} = (0, 0, 0, 0, 0, 0, 1, 0)$$

No, because

Prob 8 - Heat eq

$$u_t - u_{xx} = \sin(3\pi x)$$

$$a) \quad u_n(x, t) = e^{-\omega_n^2 t} \sin(\omega_n x)$$

Since $e^{-\omega_n^2 t}$ never will be zero we need $\sin(\omega_n x) = 0$. When $x = 0$ $\sin(x) = 0$. When $x = 1$

$$\sin(\omega_n) = 0 \quad \omega_n = n\pi$$

$$\frac{u_n}{\partial t} = -\omega_n^2 e^{-\omega_n^2 t} \sin(\omega_n x)$$

$$\frac{u_n}{\partial x} = \omega_n e^{-\omega_n^2 t} \cos(\omega_n x) \Rightarrow \frac{u_n}{\partial x^2} = -\omega_n^2 e^{-\omega_n^2 t} \sin(\omega_n x), \quad \frac{u_n}{\partial t} = \frac{u_n}{\partial x^2} \Delta.$$

$$b) \quad u(x, t) = \frac{1}{9\pi^2} \sin(3\pi x)$$

$$u(0, t) = u(1, t) = 0 \Rightarrow OK!$$

$$u_t(x, t) = 0$$

$$u_x(x, t) = \frac{1}{3\pi} \cos(3\pi x)$$

$$u_{xx}(x, t) = -\sin(3\pi x)$$

$$u_t - u_{xx} = \sin(3\pi x)$$

$$c) \quad u(x, t) = \frac{1}{9\pi^2} \sin(3\pi x) + \sum_{n=1}^{\infty} B_n e^{-\omega_n t} \sin(\omega_n x)$$

$$u(x, 0) = \frac{1}{9\pi^2} \sin(3\pi x) + \sum_{n=1}^{\infty} B_n \sin(\omega_n x) = 0 \Leftrightarrow B_n = -\frac{1}{9\pi^2}, n=3$$

Prob 9 - separation of variables

$$u_{tt} + u_{xxxx} = 0, \quad x \in [0, 1], \quad t \geq 0$$

$$a) \quad u(x, t) = F(x)G(t)$$

$$F(x)G''(t) = c F^{(4)}(x)G(t)$$

$$\frac{F^{(4)}(x)}{F(x)} = \frac{G''(t)}{cG(t)} = k$$

$$F^{(4)} - kF(x) = 0$$

$$G''(t) - ckG(t), \quad c = -1 \Rightarrow G''(t) + kG(t) = 0$$

$$b) \quad F(x) = A \sin(\beta x) + B \cos(\beta x)$$

$$F'(x) = \beta A \cos(\beta x) - \beta B \sin(\beta x)$$

$$F''(x) = -\beta^2 A \sin(\beta x) - \beta^2 B \cos(\beta x)$$

$$F^{(3)}(x) = -\beta^3 A \cos(\beta x) + \beta^3 B \sin(\beta x)$$

$$F^{(4)}(x) = \beta^4 A \sin(\beta x) + \beta^4 B \cos(\beta x)$$

$$F^{(4)} - kF(x) = 0 \Rightarrow$$

$$\beta^4 (A \sin(\beta x) + B \cos(\beta x)) - k (A \sin(\beta x) + B \cos(\beta x)) = 0, \quad k = \beta^4$$

